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- **Problem Statement ID –** SIH26223
- **Problem Statement Title-** Student Innovation
- **Theme-** Disaster Management
- **PS Category-** Hardware
- **Team ID-**
- **Team Name-** B.J.P (Bug Janta Party)



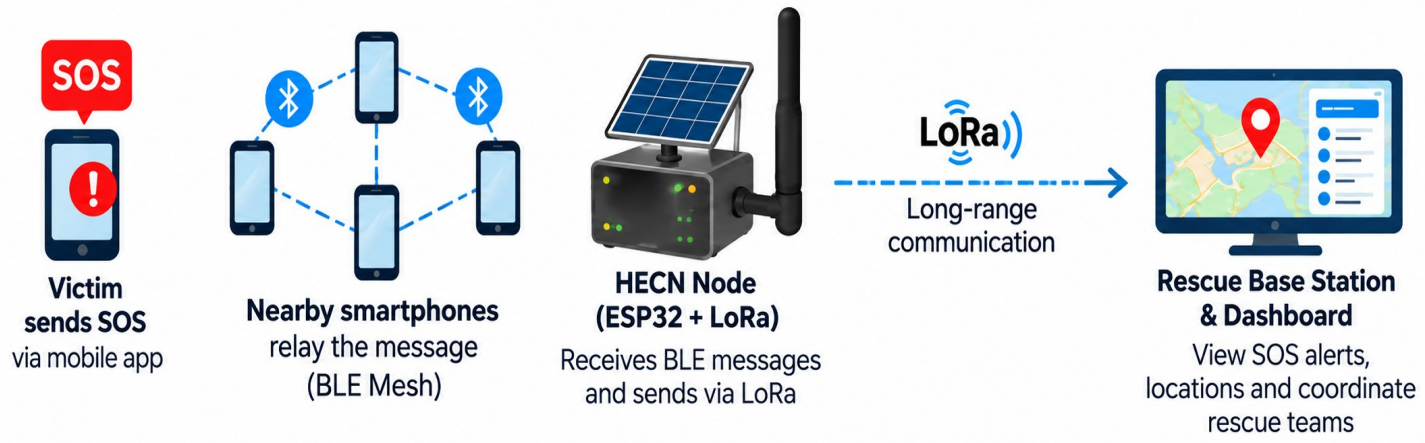
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HECN – Lifeline in Blackouts

Infrastructure – Independent Emergency Communication Network

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How HECN Works?



Infrastructure Independent

- Works when cellular networks, internet and grid power are unavailable.
- Designed for disaster and blackout

Dual-Layer Communication

- BLE (Short-Range): uses nearby participating devices to relay SOS messages
- LoRa (Long-Range): enables long-range transmission to rescue base station

THE PROBLEM

- Network outages isolate survivors
- Rescuers lack real-time SOS information
- Conventional communication systems may fail during disasters and blackout

OUR SOLUTION - HECN

- Participating smartphones relay SOS messages using BLE to a HECN node
- LoRa carries compact messages onward to a rescue base
- Solar and battery power reduce dependence on the grid

WHY THIS APPROACH

- Cost effective and easy too deploy
- Increases coverage and redundancy
- Combines short-range BLE with long-range LoRa

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TECHNICAL APPROACH

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- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts/Images/ working prototype)

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FEASIBILITY AND VIABILITY

A practical low-cost and scalable solution for real-world deployment

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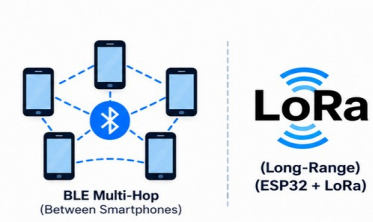
❖ Hardware Feasibility

- Uses commercially available components (ESP32, SX1278 LoRa)
- BLE-enabled smartphones act as relay devices
- Solar panel + LiFePO4 battery for off-grid operation
- Target prototype node cost: < ₹2,000 (BOM)
- Rugged and modular design for easy deployment and maintenance



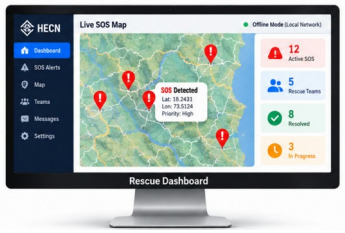
❖ Communication Feasibility

- BLE for local multi-hop communication between smartphones
- LoRa for long-range, low-bandwidth communication
- Small, optimized SOS packets (low data size)
- Message ID + TTL to avoid duplicate forwarding
- Store-and-forward mechanism for intermittent connectivity



❖ Operational/ Mission Feasibility

- Deployable as fixed HECN nodes in disaster-prone areas
- Rescue base can operate on a local/offline network
- Dashboard maps SQS locations and helps in team dispatch
- Works even when cellular networks and internet are down
- Designed for real-world harsh environments



❖ Deployment and Scale

- Prototype → Pilot deployment → Large-scale roll-out Locally available and repairable components
- Solar-powered nodes reduce dependence on grid electricity Modular design for quick installation
- Scalable Deployment (from pilot to large scale)
- Can be deployed in villages, hills, forests and disaster-hit zones



Challenges and Strategies to Overcome Them

Challenge/Risk	Engineering Fix/Strategy
Limited BLE range (50-100m)	Multi-hop relay through Participating smartphones
Intermittent connectivity (device leaves network)	Store-and-forward mechanism With message buffering
Node battery depletion (no grid power)	Solar charging + LiFePO4 battery (low-power design)
Lora packet loss (in harsh conditions)	Acknowledgement (ACK) + Retry mechanism
Hardware damage (weather, physical)	Rugged, weather-resistant Enclosure (IP-rated)

Validation Plan

Prototype (Initial build) → Lab Testing (Functionality) → Field plot (Real environment) → Scale Up (Wider deployment)

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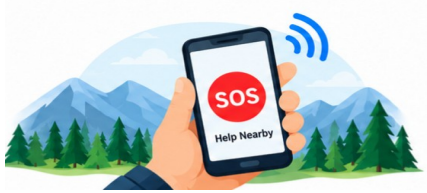
IMPACT AND BENEFITS

Maintaining a communication lifeline when conventional network fails

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For Survivors (Affected People)

- Send SOS even when cellular networks are unavailable
- Share available location information
- Messages can propagate through nearby devices
- Increases chance of being found and rescued



For Rescue Teams

- Receive consolidated SOS alerts
- View incident locations on a local dashboard
- Filter and prioritize emergency cases
- Track acknowledgement and response status
- Helps in faster and more efficient rescue operations



Broader Impact

- Improves communication in disaster and remote areas
- Low-cost and scalable solution using existing devices
- Reduces dependence on conventional infrastructure
- Supports safer and more resilient communities



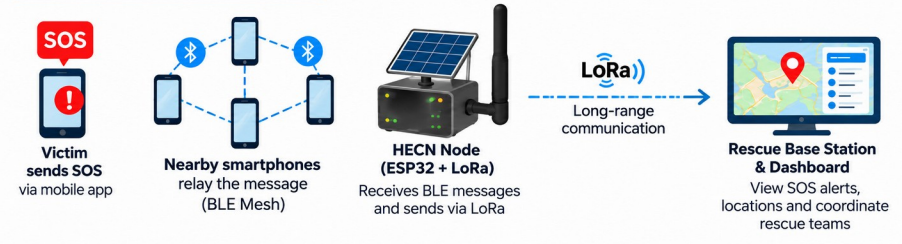
Kev Use Cases



Conventional Network during Disaster



HECN – Works Even When Networks Fail



A Safer, More Connected Tomorrow

HECN turns collective devices into a resilient communication network, saving lives when it matters most.

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RESEARCH AND REFERENCES

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- **Bluetooth SIG**: “Mesh Profile Specification 1.0.1”
  https://www.Bluetooth.com/specifications/mesh-specifications/
- **LoRa Alliance**: "LoRaWAN Link Layer Specification v1.0.4"
  https://lora-alliance.org/resource-hub/
- **NDMA**: "National Disaster Management Plan (2019)"
  https://ndma.gov.in/
- **Case Study**: "Communication Failure Analysis: Kerala Floods (2018)"
- **Google Gemini**: "For review and summary"

IMPORTANT INSTRUCTIONS

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Please ensure below pointers are met while submitting the Idea PPT:

1. **Kindly keep the maximum slides limit up to six (6). (Including the title slide)**
2. **Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures**
3. **Keep your explanation precise and easy to understand**
4. **Idea should be unique and novel.**
5. **You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).**
6. **You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.**

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.